

MICHAEL TAWFIK

Computer Science BSc Student | Full-Stack & Cloud Internship Candidate

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Availability: Summer 2026 internship · open to part-time student work during the semester

Professional Summary

Computer Science student at ELTE in Budapest with hands-on experience building full-stack and API-driven web applications using React, TypeScript, Node.js, Laravel, databases, Git, Docker, and CI/CD basics. I focus on writing maintainable code, understanding systems end-to-end, and contributing to production-minded development in a full-stack, software engineering, or cloud internship.

Work Experience

Full-Stack Developer Intern

Jun 2025 – Aug 2025

Caresynq Business Consulting

Cairo, Egypt

- **Built** React UI features for healthcare web applications used by clinic staff and partner teams.
- **Developed and maintained** REST APIs using Node.js and Express for authentication, patient records, and clinic operations.
- **Wrote and adjusted** MongoDB queries for frequently accessed collections during backend testing and debugging.
- **Helped migrate** legacy ASP.NET components to a React + Node.js stack and supported testing during rollout.
- **Worked** in an Agile team using Git-based pull requests, code reviews, daily stand-ups, and sprint planning.

Education

BSc Computer Science — GPA 4.04 / 5.0

Sep 2024 – Jul 2027

Eötvös Loránd University (ELTE)

Budapest, Hungary

- Relevant coursework: Web Programming, Databases, Software Technology, Operating Systems, Cybersecurity, Algorithms, Functional Programming.

Business Informatics & Digital Transformation — GPA 3.5 / 4.0

Sep 2023 – Aug 2024

Helwan National University

Cairo, Egypt

Projects

Customer Support / Ticketing SaaS

github.com/MichaelI402/Customer-Support-Ticketing-SaaS

Full-stack B2B support platform built as a pnpm monorepo with a Next.js 15 / React 19 frontend and a NestJS + PostgreSQL backend, delivered across 5 milestones (M0 – M4).

- Designed role-based access control for 4 user types (Customer, Agent, Manager, Admin) with JWT authentication in an httpOnly cookie and server-side privacy filters.
- Built a real-time notification system with a Socket.IO gateway and a Redis-backed BullMQ queue using idempotent jobs, so the REST API never blocks on background failures.
- Modelled the full ticketing domain in Prisma + PostgreSQL and added Vitest + Playwright tests in a GitHub Actions CI pipeline.

Tech: Next.js 15, React 19, TypeScript, NestJS, PostgreSQL, Prisma, Redis, Socket.IO, MinIO, Docker, GitHub Actions

Movie Search App

github.com/MichaelI402/movie-react-app

React app integrating the TMDB API with an Appwrite backend to track and rank trending movie searches.

- Implemented debounced search with a custom React hook to cut unnecessary API calls and improve responsiveness.
- Built trending-search analytics on Appwrite, incrementing counts per query and ranking results by popularity.
- Handled API errors, loading states, and edge cases, and secured API keys with environment variables kept out of version control.

Tech: React, Vite, JavaScript (ES6+), Appwrite, TMDB API, CSS

Face Recognition Attendance with Emotion Detection

github.com/MichaelI402/face-recognition-attendance

Real-time attendance system combining face recognition with emotion detection, built during an Intro to AI & Computer Vision summer program.

- Built a webcam pipeline in Python + OpenCV that detects and recognizes faces and logs attendance automatically to SQLite and CSV.
- Used ML / CNN-based emotion classification models to detect happy, sad, and neutral expressions in real time.
- Structured the codebase with OOP principles across the vision, classification, and data layers.

Tech: Python, OpenCV, TensorFlow / Keras, Scikit-learn, NumPy, Pandas, SQLite

Skills & Competencies

Programming Languages TypeScript · JavaScript (ES6+) · Python · Java · C · PHP · SQL · HTML · CSS

Frontend React · Next.js · TanStack Query · React Hook Form · Zod · Tailwind CSS · Vite

Backend Node.js · Express.js · NestJS · Laravel · REST APIs · Prisma ORM · JWT / RBAC · WebSocket (Socket.IO)

Databases PostgreSQL · MongoDB · MySQL · SQLite · Redis

Cloud & DevOps Docker · Docker Compose · Linux · AWS basics · Azure basics · CI/CD · GitHub Actions · Bash scripting

Tools & Testing Git / GitHub · Vitest · Playwright · JUnit 5 · ESLint · Prettier · Appwrite

Languages

English — C1 | **Arabic** — Native | **German** — A2 | **Hungarian** — A1

Certifications

- CS50: Introduction to Computer Science — Harvard University (edX)
- Problem Solving (Basic & Intermediate), C — HackerRank
- Introduction to AI & Computer Vision — Pixels Egypt Organization
- Introduction to Jenkins — Linux Foundation
- Introduction to Git & GitHub — Le Wagon
- DevOps 101 · Introduction to Linux · Git Training · Getting Started with Jenkins — Simplilearn